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# Chapter 21 - The Laplace Transform

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The previous chapters studied differential equations directly in the original independent variable. The Laplace transform introduces a different viewpoint: it converts a function of time into a function of a new variable  $s$ .

For a suitable function  $f(t)$ , write:

$$\mathcal{L}\{f(t)\} = F(s).$$

This chapter develops the **forward** transform. We will:

- define it as an improper integral
- track the values of  $s$  for which that integral converges
- derive a small table of standard transforms
- use linearity, shifting, differentiation, scaling, and division properties
- transform finite piecewise and periodic functions

Inverse transforms begin in Chapter 22. The examples, figures, derivations, and exercises here are independently constructed; the reference chapter is used only as a topic map.

## From A Time Function To An S-Function

### The Goal Of This Block

The goal is to understand every symbol in the Laplace-transform definition and why convergence must be checked rather than assumed.

### The Definition

For a function  $f(t)$  defined for  $t \geq 0$ , its Laplace transform is:

$$F(s) = \mathcal{L}\{f(t)\} = \int_0^{\infty} e^{-st} f(t) dt .$$

Here:

- $t$  is the integration variable, usually representing time
- $s$  is a parameter and is treated as constant during the integration
- $f(t)$  is the original function
- $F(s)$  is the transformed function
- $e^{-st}$  is the transform kernel

The braces in  $\mathcal{L}\{f(t)\}$  identify the function being transformed. They are not multiplication symbols.

### The Integral Is Improper

The upper limit  $\infty$  means that the definition is shorthand for a limit:

$$\mathcal{L}\{f(t)\} = \lim_{R \rightarrow \infty} \int_0^R e^{-st} f(t) dt .$$

The transform exists at a particular  $s$  only when this limit is finite.

A formula for  $F(s)$  without its convergence condition is incomplete. The same function can converge for some  $s$  and diverge for others.

## What The Kernel Does

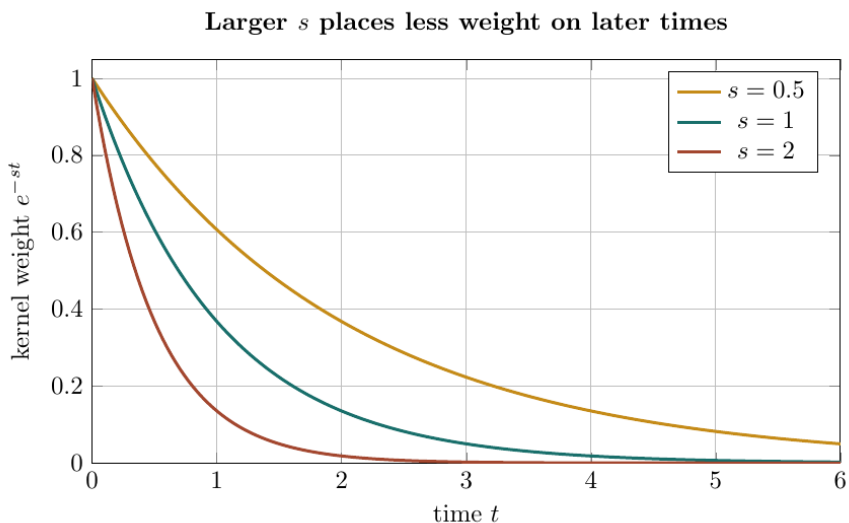


Figure 1: Exponential kernels for three positive values of  $s$

For  $s > 0$ , the kernel  $e^{-st}$  decays as  $t$  increases. A larger  $s$  makes it decay faster, so late-time values of  $f(t)$  receive less weight.

The key intuition:

**$F(s)$  is a weighted summary of the entire future half-line  $t \geq 0$ , and changing  $s$  changes how strongly later times contribute.**

## A First Transform From The Definition

Find the transform of:

$$f(t) = e^{-3t}.$$

Substitute  $f(t)$  into the definition:

$$\begin{aligned}\mathcal{L}\{e^{-3t}\} &= \int_0^{\infty} e^{-st} e^{-3t} dt \\ &= \int_0^{\infty} e^{-(s+3)t} dt.\end{aligned}$$

Replace the improper integral by a limit:

$$\mathcal{L}\{e^{-3t}\} = \lim_{R \rightarrow \infty} \int_0^R e^{-(s+3)t} dt.$$

For  $s + 3 \neq 0$ , integrate with respect to  $t$ :

$$\begin{aligned}\int_0^R e^{-(s+3)t} dt &= \left[ -\frac{e^{-(s+3)t}}{s+3} \right]_0^R \\ &= -\frac{e^{-(s+3)R}}{s+3} + \frac{1}{s+3}.\end{aligned}$$

The exponential term tends to zero exactly when:

$$s + 3 > 0,$$

or equivalently:

$$s > -3.$$

Since the remainder  $e^{-(s+3)R}/(s+3)$  vanishes when  $s > -3$ , taking  $R \rightarrow \infty$  gives:

$$\boxed{\mathcal{L}\{e^{-3t}\} = \frac{1}{s+3}, \quad s > -3.}$$

### Why The Other S-Values Fail

If  $s = -3$ , then the integrand becomes 1:

$$\int_0^R 1 \, dt = R,$$

which diverges as  $R \rightarrow \infty$ .

If  $s < -3$ , then  $-(s+3) > 0$ , so  $e^{-(s+3)t}$  grows rather than decays. The integral again diverges.

Thus the condition  $s > -3$  follows from the limiting process; it is not an optional note attached after the calculation.

### A Practical Existence Test

A common sufficient condition is:

- $f$  is piecewise continuous on every finite interval  $[0, R]$
- there are constants  $M > 0$ ,  $a$ , and  $T$  such that

$$|f(t)| \leq Me^{at} \quad \text{for } t \geq T$$

Then  $f$  is said to be of **exponential order**  $a$ , and its Laplace transform converges for sufficiently large  $s$ , normally for  $s > a$ .

This condition is sufficient, not a claim that every transform has the same largest possible convergence region.

## Block 1 Summary

The Laplace transform is an improper weighted integral:

$$F(s) = \int_0^{\infty} e^{-st} f(t) dt.$$

Treat  $s$  as a parameter, integrate with respect to  $t$ , evaluate the limit, and state the values of  $s$  for which it exists.

## Building The Core Transform Table

### The Goal Of This Block

The goal is to derive the transforms of constants, powers, and exponentials directly from the definition.

### The Transform Of One

Start from the definition:

$$\mathcal{L}\{1\} = \lim_{R \rightarrow \infty} \int_0^R e^{-st} dt.$$

For  $s \neq 0$ :

$$\begin{aligned} \int_0^R e^{-st} dt &= \left[ -\frac{1}{s} e^{-st} \right]_0^R \\ &= -\frac{e^{-sR}}{s} + \frac{1}{s}. \end{aligned}$$

When  $s > 0$ ,  $e^{-sR} \rightarrow 0$ . Therefore:

$$\boxed{\mathcal{L}\{1\} = \frac{1}{s}, \quad s > 0.}$$

At  $s = 0$ , the integral is  $\int_0^{\infty} 1 dt$ , which diverges. For  $s < 0$ , the kernel grows.



## The Transform Of T

Use integration by parts on the finite interval  $[0, R]$ :

$$u = t, \quad dv = e^{-st} dt.$$

Differentiating  $u = t$  and integrating  $dv = e^{-st} dt$  gives:

$$du = dt, \quad v = -\frac{1}{s}e^{-st}.$$

Apply  $\int u dv = uv - \int v du$ :

$$\begin{aligned} \int_0^R te^{-st} dt &= \left[ -\frac{t}{s}e^{-st} \right]_0^R + \frac{1}{s} \int_0^R e^{-st} dt \\ &= -\frac{R}{s}e^{-sR} + \frac{1}{s} \left( -\frac{e^{-sR}}{s} + \frac{1}{s} \right). \end{aligned}$$

For  $s > 0$ , both  $e^{-sR}$  and  $Re^{-sR}$  tend to zero. Hence:

$$\boxed{\mathcal{L}\{t\} = \frac{1}{s^2}, \quad s > 0}.$$

## The Power Recurrence

Define:

$$I_n(s) = \int_0^\infty t^n e^{-st} dt.$$

For a positive integer  $n$ , integrate by parts with:

$$u = t^n, \quad dv = e^{-st} dt.$$

Differentiating  $u = t^n$  and integrating  $dv = e^{-st} dt$  gives:

$$du = nt^{n-1} dt, \quad v = -\frac{1}{s}e^{-st}.$$

The boundary term vanishes for  $s > 0$ , so:

$$\begin{aligned} I_n(s) &= \left[ -\frac{t^n}{s} e^{-st} \right]_0^\infty + \frac{n}{s} \int_0^\infty t^{n-1} e^{-st} dt \\ &= \frac{n}{s} I_{n-1}(s). \end{aligned}$$

Since  $I_0(s) = 1/s$ , repeated use gives:

$$\begin{aligned} I_n(s) &= \frac{n}{s} \frac{n-1}{s} \cdots \frac{1}{s} I_0(s) \\ &= \frac{n!}{s^{n+1}}. \end{aligned}$$

Combining the recurrence with  $I_0(s) = 1/s$  gives the power-transform rule:

$$\boxed{\mathcal{L}\{t^n\} = \frac{n!}{s^{n+1}}, \quad s > 0.}$$

The symbol  $n!$  means  $n(n-1) \cdots 2 \cdot 1$ .

### A Power Example

For  $t^4$ , set  $n = 4$  in the power formula:

$$\begin{aligned} \mathcal{L}\{t^4\} &= \frac{4!}{s^{4+1}} \\ &= \frac{4 \cdot 3 \cdot 2 \cdot 1}{s^5} \\ &= \frac{24}{s^5}, \quad s > 0. \end{aligned}$$

Do not confuse the exponent 4 with the factorial  $4!$  or the denominator power 5.

## The General Exponential Transform

For  $f(t) = e^{at}$ :

$$\begin{aligned}\mathcal{L}\{e^{at}\} &= \int_0^{\infty} e^{-st} e^{at} dt \\ &= \int_0^{\infty} e^{-(s-a)t} dt.\end{aligned}$$

This is the transform of 1 with  $s$  replaced by  $s - a$ . Therefore:

$$\boxed{\mathcal{L}\{e^{at}\} = \frac{1}{s - a}, \quad s > a}.$$

The sign in the denominator is opposite the sign in the exponent because:

$$e^{-st} e^{at} = e^{-(s-a)t}.$$

## A Small Core Table

The results derived so far are:

| $f(t)$   | $\mathcal{L}\{f(t)\}$ | Condition |
|----------|-----------------------|-----------|
| 1        | $1/s$                 | $s > 0$   |
| $t^n$    | $n!/s^{n+1}$          | $s > 0$   |
| $e^{at}$ | $1/(s - a)$           | $s > a$   |

The condition column belongs to the transform pair. Keep it visible when the formula is derived from an improper integral.

## Block 2 Summary

Integration by parts produces a recurrence for powers, while combining exponents gives the exponential transform. The basic table is small because later properties generate many new transforms from these entries.

## Sine, Cosine, And Hyperbolic Functions

### The Goal Of This Block

The goal is to derive the sine and cosine transforms as a coupled pair and then obtain hyperbolic transforms from exponentials.

### Define Two Linked Integrals

For  $s > 0$ , define:

$$C(s) = \int_0^{\infty} e^{-st} \cos(bt) dt$$

and:

$$S(s) = \int_0^{\infty} e^{-st} \sin(bt) dt.$$

The exponential kernel ensures convergence because sine and cosine remain bounded.

### First Integration By Parts

For  $C(s)$ , choose:

$$u = e^{-st}, \quad dv = \cos(bt) dt.$$

Differentiating  $u = e^{-st}$  and integrating  $dv = \cos(bt) dt$  gives:

$$du = -se^{-st} dt, \quad v = \frac{1}{b} \sin(bt).$$

The boundary term is zero for  $s > 0$ , so:

$$\begin{aligned} C(s) &= \left[ \frac{1}{b} e^{-st} \sin(bt) \right]_0^\infty + \frac{s}{b} \int_0^\infty e^{-st} \sin(bt) dt \\ &= \frac{s}{b} S(s). \end{aligned}$$

## Second Integration By Parts

For  $S(s)$ , choose:

$$u = e^{-st}, \quad dv = \sin(bt) dt.$$

Differentiating  $u = e^{-st}$  and integrating  $dv = \sin(bt) dt$  gives:

$$du = -se^{-st} dt, \quad v = -\frac{1}{b} \cos(bt).$$

Apply integration by parts:

$$\begin{aligned} S(s) &= \left[ -\frac{1}{b} e^{-st} \cos(bt) \right]_0^\infty - \frac{s}{b} \int_0^\infty e^{-st} \cos(bt) dt \\ &= \frac{1}{b} - \frac{s}{b} C(s). \end{aligned}$$

The boundary term equals  $1/b$  because its value at infinity is 0 and its value at  $t = 0$  is  $-1/b$ .

## Solve The Two Equations

Use:

$$C(s) = \frac{s}{b}S(s)$$

inside:

$$S(s) = \frac{1}{b} - \frac{s}{b}C(s).$$

Substitute and simplify one step at a time:

$$\begin{aligned} S(s) &= \frac{1}{b} - \frac{s}{b} \left( \frac{s}{b} S(s) \right) \\ &= \frac{1}{b} - \frac{s^2}{b^2} S(s). \end{aligned}$$

Add  $(s^2/b^2)S(s)$  to both sides:

$$\left( 1 + \frac{s^2}{b^2} \right) S(s) = \frac{1}{b}.$$

Write the factor with one denominator:

$$\frac{s^2 + b^2}{b^2} S(s) = \frac{1}{b}.$$

Multiply by  $b^2/(s^2 + b^2)$ :

$$S(s) = \frac{b}{s^2 + b^2}.$$

Substitute the solved value of  $S(s)$  into  $C(s) = (s/b)S(s)$ :

$$\begin{aligned} C(s) &= \frac{s}{b} \frac{b}{s^2 + b^2} \\ &= \frac{s}{s^2 + b^2}. \end{aligned}$$

The solved functions  $S(s)$  and  $C(s)$  give the two transform formulas:

$$\boxed{\mathcal{L}\{\sin(bt)\} = \frac{b}{s^2 + b^2}, \quad s > 0}$$

and:

$$\boxed{\mathcal{L}\{\cos(bt)\} = \frac{s}{s^2 + b^2}, \quad s > 0}.$$

### How To Remember The Numerators

The denominators match:

$$s^2 + b^2.$$

The sine transform has  $b$  in the numerator, while the cosine transform has  $s$ :

$$\sin(bt) \rightarrow \frac{b}{s^2 + b^2}, \quad \cos(bt) \rightarrow \frac{s}{s^2 + b^2}.$$

Check dimensions or set  $b = 0$ :  $\cos(0t) = 1$  should give  $1/s$ , which confirms the cosine numerator.

## Hyperbolic Functions From Exponentials

Use:

$$\cosh(bt) = \frac{e^{bt} + e^{-bt}}{2}.$$

Applying the exponential transform and combining fractions gives:

$$\begin{aligned}\mathcal{L}\{\cosh(bt)\} &= \frac{1}{2} \left( \frac{1}{s-b} + \frac{1}{s+b} \right) \\ &= \frac{1}{2} \left( \frac{(s+b) + (s-b)}{s^2 - b^2} \right) \\ &= \frac{s}{s^2 - b^2}.\end{aligned}$$

Similarly, because:

$$\sinh(bt) = \frac{e^{bt} - e^{-bt}}{2},$$

we obtain:

$$\mathcal{L}\{\sinh(bt)\} = \frac{b}{s^2 - b^2}.$$

For real  $b$ , both formulas converge for  $s > |b|$ .

### Block 3 Summary

Sine and cosine form a linked integration-by-parts system. Hyperbolic sine and cosine follow from their exponential definitions. The signs in  $s^2 + b^2$  and  $s^2 - b^2$  distinguish circular and hyperbolic functions.



## Linearity And Direct Piecewise Transforms

### The Goal Of This Block

The goal is to transform combinations term by term and handle a finite piecewise function directly from the integral definition.

### Linearity

If:

$$\mathcal{L}\{f(t)\} = F(s), \quad \mathcal{L}\{g(t)\} = G(s),$$

then for constants  $a$  and  $b$ :

$$\boxed{\mathcal{L}\{af(t) + bg(t)\} = aF(s) + bG(s)}.$$

This follows from linearity of integration:

$$\begin{aligned} \int_0^{\infty} e^{-st} [af(t) + bg(t)] dt &= a \int_0^{\infty} e^{-st} f(t) dt \\ &\quad + b \int_0^{\infty} e^{-st} g(t) dt. \end{aligned}$$

### A Complete Linearity Example

Find:

$$\mathcal{L}\{2t^3 - 5\sin(4t) + 7e^{-2t}\}.$$

Apply linearity before substituting table entries:

$$\begin{aligned} \mathcal{L}\{2t^3 - 5\sin(4t) + 7e^{-2t}\} &= 2\mathcal{L}\{t^3\} - 5\mathcal{L}\{\sin(4t)\} \\ &\quad + 7\mathcal{L}\{e^{-2t}\}. \end{aligned}$$

Using the power, sine, and exponential rows of the transform table gives:

$$\mathcal{L}\{t^3\} = \frac{3!}{s^4} = \frac{6}{s^4},$$

$$\mathcal{L}\{\sin(4t)\} = \frac{4}{s^2 + 16},$$

and:

$$\mathcal{L}\{e^{-2t}\} = \frac{1}{s + 2}.$$

Substitute these three results:

$$\boxed{\mathcal{L}\{2t^3 - 5\sin(4t) + 7e^{-2t}\} = \frac{12}{s^4} - \frac{20}{s^2 + 16} + \frac{7}{s + 2}}.$$

All three terms converge together for  $s > 0$ , the most restrictive of their individual conditions.

### A Finite Piecewise Function

Let:

$$q(t) = \begin{cases} t, & 0 \leq t < 2, \\ 0, & t \geq 2. \end{cases}$$

Because  $q(t) = 0$  after  $t = 2$ , the transform reduces to a finite integral:

$$\mathcal{L}\{q(t)\} = \int_0^2 te^{-st} dt.$$

For this integrand, integration by parts gives the antiderivative:

$$\int te^{-st} dt = -\frac{t}{s}e^{-st} - \frac{1}{s^2}e^{-st}.$$

Evaluate at both bounds:

$$\begin{aligned}\mathcal{L}\{q(t)\} &= \left[ -\frac{t}{s}e^{-st} - \frac{1}{s^2}e^{-st} \right]_0^2 \\ &= -\frac{2e^{-2s}}{s} - \frac{e^{-2s}}{s^2} + \frac{1}{s^2} \\ &= \frac{1 - e^{-2s}(1 + 2s)}{s^2}.\end{aligned}$$

Thus, for  $s \neq 0$ :

$$\boxed{\mathcal{L}\{q(t)\} = \frac{1 - e^{-2s}(1 + 2s)}{s^2}}.$$

At  $s = 0$ , return to the finite integral rather than substituting into the fraction:

$$\mathcal{L}\{q(t)\}|_{s=0} = \int_0^2 t dt = 2.$$

The apparent singularity at  $s = 0$  is removable.

### Common Linearity Mistakes

Linearity permits sums and constant multiples. It does not generally permit:

$$\mathcal{L}\{f(t)g(t)\} = \mathcal{L}\{f(t)\}\mathcal{L}\{g(t)\}.$$

That equation is usually false.

Also keep the whole argument of a trigonometric or exponential function visible. For example, the 4 in  $\sin(4t)$  becomes both the numerator 4 and the squared denominator term 16.

## Block 4 Summary

Linearity turns sums into sums of transforms. A piecewise function can always be handled by splitting the defining integral over the intervals on which its formula changes.

## Properties That Generate New Transforms

### The Goal Of This Block

The goal is to use one known transform  $F(s)$  to generate transforms involving exponential factors, powers of  $t$ , scaled inputs, and division by  $t$ .

### Exponential Shifting

Suppose:

$$\mathcal{L}\{f(t)\} = F(s).$$

Multiply the original function by  $e^{at}$  and return to the definition:

$$\begin{aligned}\mathcal{L}\{e^{at}f(t)\} &= \int_0^{\infty} e^{-st} e^{at} f(t) dt \\ &= \int_0^{\infty} e^{-(s-a)t} f(t) dt.\end{aligned}$$

The last integral is  $F$  evaluated at  $s - a$ :

$$\boxed{\mathcal{L}\{e^{at}f(t)\} = F(s - a)}.$$

This is a shift in the transform variable, not a shift in time.

## A Complete Shift Example

Find:

$$\mathcal{L}\{e^{2t}(t^2 + 3)\}.$$

First transform the unshifted function  $f(t) = t^2 + 3$ :

$$\begin{aligned} F(s) &= \mathcal{L}\{t^2\} + 3\mathcal{L}\{1\} \\ &= \frac{2!}{s^3} + \frac{3}{s} \\ &= \frac{2}{s^3} + \frac{3}{s}. \end{aligned}$$

The factor  $e^{2t}$  means  $a = 2$ , so replace every  $s$  in  $F(s)$  by  $s - 2$ :

$$\boxed{\mathcal{L}\{e^{2t}(t^2 + 3)\} = \frac{2}{(s-2)^3} + \frac{3}{s-2}, \quad s > 2.}$$

The convergence boundary also shifts from  $s > 0$  to  $s - 2 > 0$ .

## Multiplication By T

Differentiate the defining integral with respect to  $s$  under suitable regularity and convergence assumptions:

$$\begin{aligned} F'(s) &= \frac{d}{ds} \int_0^\infty e^{-st} f(t) dt \\ &= \int_0^\infty \frac{\partial}{\partial s} (e^{-st} f(t)) dt \\ &= \int_0^\infty (-t) e^{-st} f(t) dt. \end{aligned}$$

The differentiated integral states  $F'(s) = -\mathcal{L}\{tf(t)\}$ . Solving this identity for the transform gives:

$$\boxed{\mathcal{L}\{tf(t)\} = -F'(s)}.$$

Repeating the operation gives:

$$\boxed{\mathcal{L}\{t^n f(t)\} = (-1)^n F^{(n)}(s)}.$$

The exponent  $n$  counts derivatives with respect to  $s$ .

### A Two-Derivative Example

Find  $\mathcal{L}\{t^2 \sin(2t)\}$ .

Begin with:

$$F(s) = \mathcal{L}\{\sin(2t)\} = \frac{2}{s^2 + 4}.$$

Because  $n = 2$ :

$$\mathcal{L}\{t^2 \sin(2t)\} = F''(s).$$

Differentiate once:

$$\begin{aligned} F'(s) &= 2(-1)(s^2 + 4)^{-2}(2s) \\ &= -\frac{4s}{(s^2 + 4)^2}. \end{aligned}$$

Differentiate the product  $-4s(s^2 + 4)^{-2}$ :

$$\begin{aligned} F''(s) &= -4(s^2 + 4)^{-2} + (-4s)(-2)(s^2 + 4)^{-3}(2s) \\ &= -\frac{4}{(s^2 + 4)^2} + \frac{16s^2}{(s^2 + 4)^3}. \end{aligned}$$

Rewrite the first fraction over the common denominator  $(s^2 + 4)^3$ :

$$\begin{aligned} F''(s) &= \frac{-4(s^2 + 4) + 16s^2}{(s^2 + 4)^3} \\ &= \frac{12s^2 - 16}{(s^2 + 4)^3} \\ &= \frac{4(3s^2 - 4)}{(s^2 + 4)^3}. \end{aligned}$$

Thus:

$$\boxed{\mathcal{L}\{t^2 \sin(2t)\} = \frac{4(3s^2 - 4)}{(s^2 + 4)^3}, \quad s > 0.}$$

### Scaling The Independent Variable

Let  $c > 0$ . Start from:

$$\mathcal{L}\{f(ct)\} = \int_0^\infty e^{-st} f(ct) dt.$$

Use the substitution:

$$u = ct, \quad t = \frac{u}{c}, \quad dt = \frac{du}{c}.$$

After the substitution  $u = ct$ , the transform integral becomes:

$$\begin{aligned}\mathcal{L}\{f(ct)\} &= \frac{1}{c} \int_0^\infty e^{-(s/c)u} f(u) du \\ &= \frac{1}{c} F\left(\frac{s}{c}\right).\end{aligned}$$

This change of variable proves the scaling rule:

$$\boxed{\mathcal{L}\{f(ct)\} = \frac{1}{c} F\left(\frac{s}{c}\right), \quad c > 0.}$$

Both the outside factor  $1/c$  and the replacement  $s \mapsto s/c$  are needed.

### Division By T

If  $F(s) = \mathcal{L}\{f(t)\}$  and the required integrals converge, then:

$$\boxed{\mathcal{L}\left\{\frac{f(t)}{t}\right\} = \int_s^\infty F(u) du.}$$

The variable  $u$  is a dummy transform variable. It prevents confusion with the lower limit  $s$ .

As an example, use:

$$\mathcal{L}\{\sin(5t)\} = \frac{5}{s^2 + 25}.$$

Insert  $F(u) = 5/(u^2 + 25)$  into the division-by- $t$  formula:

$$\begin{aligned}\mathcal{L}\left\{\frac{\sin(5t)}{t}\right\} &= \int_s^\infty \frac{5}{u^2 + 25} du \\ &= \left[ \arctan\left(\frac{u}{5}\right) \right]_s^\infty \\ &= \frac{\pi}{2} - \arctan\left(\frac{s}{5}\right).\end{aligned}$$



For  $s > 0$ , the complementary-angle identity gives:

$$\mathcal{L} \left\{ \frac{\sin(5t)}{t} \right\} = \arctan \left( \frac{5}{s} \right).$$

Although the expression contains  $1/t$ , the original function has a finite limit 5 as  $t \rightarrow 0$ .

### Block 5 Summary

The four property patterns are:

$$\begin{aligned} e^{at}f(t) &\longrightarrow F(s-a), \\ t^n f(t) &\longrightarrow (-1)^n F^{(n)}(s), \\ f(ct) &\longrightarrow \frac{1}{c} F(s/c), \\ \frac{f(t)}{t} &\longrightarrow \int_s^\infty F(u) du. \end{aligned}$$

Each property changes a known transform in a specific way. Do not combine the rules by visual guesswork.

## Periodic Functions

### The Goal Of This Block

The goal is to reduce an integral over infinitely many repeated periods to one integral over a single period.

### The Periodic Transform Formula

Suppose  $p(t)$  has period  $T > 0$ :

$$p(t + T) = p(t).$$

Split the transform into period-length intervals:

$$\mathcal{L}\{p(t)\} = \sum_{k=0}^{\infty} \int_{kT}^{(k+1)T} e^{-st} p(t) dt.$$

In the  $k$ th interval, set  $t = u + kT$ . Periodicity gives  $p(u + kT) = p(u)$ , so:

$$\begin{aligned} \int_{kT}^{(k+1)T} e^{-st} p(t) dt &= \int_0^T e^{-s(u+kT)} p(u) du \\ &= e^{-skT} \int_0^T e^{-su} p(u) du. \end{aligned}$$

Factor out the one-period integral:

$$\mathcal{L}\{p(t)\} = \left( \int_0^T e^{-su} p(u) du \right) \sum_{k=0}^{\infty} e^{-skT}.$$

For  $s > 0$ , the geometric series is:

$$\sum_{k=0}^{\infty} e^{-skT} = \frac{1}{1 - e^{-sT}}.$$

Since the repeated periods contribute the geometric factor  $\sum_{k=0}^{\infty} e^{-skT} = 1/(1 - e^{-sT})$ , multiplying by the one-period integral gives:

$$\mathcal{L}\{p(t)\} = \frac{\int_0^T e^{-st} p(t) dt}{1 - e^{-sT}}.$$

### An Original Periodic Ramp

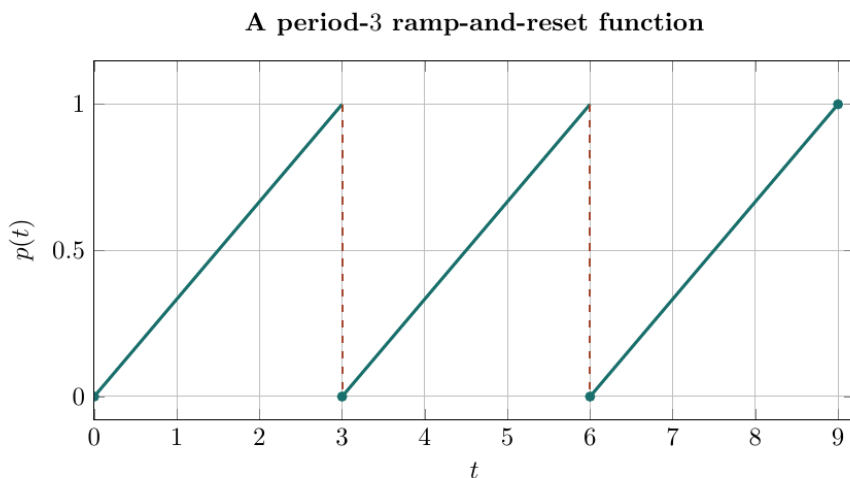


Figure 2: A period-three ramp-and-reset function

Define one period by:

$$p(t) = \frac{t}{3}, \quad 0 \leq t < 3,$$

and repeat it with period  $T = 3$ .

The jumps at  $t = 3, 6, 9, \dots$  do not prevent the transform from existing. The function is piecewise continuous and bounded.

## Transform The Periodic Ramp

Use the periodic formula with  $T = 3$ :

$$\mathcal{L}\{p(t)\} = \frac{\int_0^3 e^{-st} \frac{t}{3} dt}{1 - e^{-3s}}.$$

Factor out  $1/3$ :

$$\int_0^3 e^{-st} \frac{t}{3} dt = \frac{1}{3} \int_0^3 te^{-st} dt.$$

Use:

$$\int te^{-st} dt = -\frac{t}{s}e^{-st} - \frac{1}{s^2}e^{-st}.$$

Evaluate from 0 to 3:

$$\begin{aligned} \frac{1}{3} \int_0^3 te^{-st} dt &= \frac{1}{3} \left[ -\frac{t}{s}e^{-st} - \frac{1}{s^2}e^{-st} \right]_0^3 \\ &= \frac{1}{3} \left( -\frac{3e^{-3s}}{s} - \frac{e^{-3s}}{s^2} + \frac{1}{s^2} \right) \\ &= \frac{1 - e^{-3s}(1 + 3s)}{3s^2}. \end{aligned}$$

Substitute this one-period integral into the periodic formula:

$$\boxed{\mathcal{L}\{p(t)\} = \frac{1 - e^{-3s}(1 + 3s)}{3s^2(1 - e^{-3s})}, \quad s > 0.}$$

## Why One Period Is Enough

Every later period has the same shape, but the kernel contributes an extra factor:

$$1, \quad e^{-sT}, \quad e^{-2sT}, \quad e^{-3sT}, \quad \dots$$

The denominator  $1 - e^{-sT}$  is the sum of those repeated exponential weights. It does not come from integrating the one-period formula itself.

## Common Periodic-Function Mistakes

Do not:

- use the wrong period in  $e^{-sT}$
- integrate over several periods and still divide by  $1 - e^{-sT}$
- forget a piece of the one-period definition
- assume a jump discontinuity prevents a transform
- replace  $p(t)$  by its formula from the first period outside that period without declaring periodic extension

## Block 6 Summary

For a period- $T$  function, transform one complete period and divide by the geometric-series factor  $1 - e^{-sT}$ . The numerator contains the shape; the denominator accounts for repetition.

# A Reliable Transform Workflow

## The Goal Of This Block

The goal is to choose the simplest valid route and audit the result before it is used in later differential-equation work.

## Choose The Route Before Calculating

Ask these questions in order:

- 1. Is the function already a basic table entry?
- 2. Is it a sum or constant multiple suited to linearity?
- 3. Does it contain  $e^{at}$ ,  $t^n$ ,  $f(ct)$ , or  $f(t)/t$ ?
- 4. Is it piecewise with finite support?
- 5. Is it periodic?
- 6. If none applies cleanly, should the definition be used directly?

This prevents a short property calculation from being replaced by several pages of unnecessary integration.

## Core Table For This Chapter

| $f(t)$      | $F(s) = \mathcal{L}\{f(t)\}$ |
|-------------|------------------------------|
| 1           | $1/s$                        |
| $t^n$       | $n!/s^{n+1}$                 |
| $e^{at}$    | $1/(s - a)$                  |
| $\sin(bt)$  | $b/(s^2 + b^2)$              |
| $\cos(bt)$  | $s/(s^2 + b^2)$              |
| $\sinh(bt)$ | $b/(s^2 - b^2)$              |
| $\cosh(bt)$ | $s/(s^2 - b^2)$              |

The table is a starting point. Conditions of convergence still follow from the original function and any shifts that have been applied.

## Fast Sanity Checks

Before accepting a transform, check:

- **Linearity:** were all outside constants preserved?
- **Power:** did  $t^n$  produce  $n!$  and denominator power  $n + 1$ ?
- **Trigonometry:** does sine have  $b$  and cosine have  $s$  in the numerator?
- **Shift:** did  $e^{at}$  produce  $s - a$ , not  $s + a$ ?
- **Scaling:** are both  $1/c$  and  $s/c$  present?
- **Periodicity:** is the denominator  $1 - e^{-sT}$  using the true period?
- **Convergence:** is the stated  $s$ -range compatible with growth of  $f(t)$ ?

## Block 7 Summary

Identify structure before calculating, apply one property at a time, and keep the original time function visible while checking signs, parameters, and the convergence region.

## Original Practice Set

### Problems 1 To 4: Definition And Core Table

**Problem 1: Transform A Constant From The Definition.** Use the improper-integral definition to find  $\mathcal{L}\{7\}$  and state its convergence condition.

**Problem 2: Find A Convergence Boundary.** Use the definition to transform  $e^{4t}$  and determine exactly which real values of  $s$  give convergence.

**Problem 3: Transform A Power.** Find  $\mathcal{L}\{t^5\}$  and expand the factorial.

**Problem 4: Use Linearity.** Find:

$$\mathcal{L}\{3t^2 - 2\sin(5t)\}.$$

**Problems 5 To 8: Transform Properties**

**Problem 5: Apply An Exponential Shift.** Find  $\mathcal{L}\{e^{-3t}(t+2)\}$  and state a convergence condition.

**Problem 6: Multiply By T.** Use differentiation in  $s$  to find  $\mathcal{L}\{te^{-2t}\}$ .

**Problem 7: Differentiate Twice.** Find  $\mathcal{L}\{t^2 \sin(2t)\}$  without integrating in  $t$ .

**Problem 8: Scale The Input.** Given  $f(t) = \sin t$  and  $F(s) = 1/(s^2 + 1)$ , use the scaling property to find  $\mathcal{L}\{f(3t)\}$ .

**Problems 9 To 12: Division, Pieces, And Periodicity**

**Problem 9: Divide By T.** Use the division property to find  $\mathcal{L}\{\sin(5t)/t\}$  for  $s > 0$ .

**Problem 10: Transform A Finite Pulse.** Let:

$$r(t) = \begin{cases} 1, & 0 \leq t < 2, \\ 0, & t \geq 2. \end{cases}$$

Find  $\mathcal{L}\{r(t)\}$  directly from the definition.

**Problem 11: Transform A Periodic Pulse.** A period-3 function equals 2 on  $0 \leq t < 1$  and 0 on  $1 \leq t < 3$ . Find its Laplace transform.

**Problem 12: Diagnose A Shift Error.** A learner writes:

$$\mathcal{L}\{e^{3t}f(t)\} = F(s+3).$$

Explain the sign error, give the correct formula, and describe how a boundary  $s > \alpha$  for  $F(s)$  changes.



**Worked Answers: Problems 1 To 4****Answer 1.** Start from the definition:

$$\begin{aligned}
 \mathcal{L}\{7\} &= \lim_{R \rightarrow \infty} \int_0^R 7e^{-st} dt \\
 &= 7 \lim_{R \rightarrow \infty} \left[ -\frac{1}{s} e^{-st} \right]_0^R \\
 &= 7 \lim_{R \rightarrow \infty} \left( -\frac{e^{-sR}}{s} + \frac{1}{s} \right).
 \end{aligned}$$

For  $s > 0$ , the exponential tends to zero. Therefore:

$$\boxed{\mathcal{L}\{7\} = \frac{7}{s}, \quad s > 0.}$$

**Answer 2.** Combine the exponentials:

$$\begin{aligned}
 \mathcal{L}\{e^{4t}\} &= \int_0^\infty e^{-st} e^{4t} dt \\
 &= \int_0^\infty e^{-(s-4)t} dt.
 \end{aligned}$$

The kernel decays when  $s - 4 > 0$ . Hence:

$$\boxed{\mathcal{L}\{e^{4t}\} = \frac{1}{s-4}, \quad s > 4.}$$

At  $s = 4$  the integrand is 1; for  $s < 4$  it grows.

**Answer 3.** Use the power formula with  $n = 5$ :

$$\begin{aligned}\mathcal{L}\{t^5\} &= \frac{5!}{s^6} \\ &= \frac{5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}{s^6} \\ &= \boxed{\frac{120}{s^6}}, \quad s > 0.\end{aligned}$$

**Answer 4.** Apply linearity and then substitute the two table entries:

$$\begin{aligned}\mathcal{L}\{3t^2 - 2\sin(5t)\} &= 3\frac{2!}{s^3} - 2\frac{5}{s^2 + 25} \\ &= \boxed{\frac{6}{s^3} - \frac{10}{s^2 + 25}}, \quad s > 0.\end{aligned}$$

**Worked Answers: Problems 5 To 8**

**Answer 5.** First transform  $f(t) = t + 2$ , obtaining  $F(s) = 1/s^2 + 2/s$ .

The factor  $e^{-3t}$  has  $a = -3$ , so replace  $s$  by  $s - (-3) = s + 3$ :

$$\mathcal{L}\{e^{-3t}(t+2)\} = \frac{1}{(s+3)^2} + \frac{2}{s+3}, \quad s > -3.$$

**Answer 6.** Start from  $F(s) = \mathcal{L}\{e^{-2t}\} = 1/(s+2)$ .

Use  $\mathcal{L}\{tf(t)\} = -F'(s)$ :

$$\begin{aligned} \mathcal{L}\{te^{-2t}\} &= -\frac{d}{ds}(s+2)^{-1} \\ &= -(-(s+2)^{-2}) \\ &= \boxed{\frac{1}{(s+2)^2}}, \quad s > -2. \end{aligned}$$

**Answer 7.** For  $f(t) = \sin(2t)$ , the base transform is  $F(s) = 2/(s^2+4)$ . Since multiplication by  $t^2$  corresponds to  $F''(s)$ :

$$\begin{aligned} F'(s) &= -\frac{4s}{(s^2+4)^2}, \\ F''(s) &= -\frac{4}{(s^2+4)^2} + \frac{16s^2}{(s^2+4)^3} \\ &= \frac{4(3s^2-4)}{(s^2+4)^3}. \end{aligned}$$

Because the required transform is  $F''(s)$ , the calculation gives:

$$\mathcal{L}\{t^2 \sin(2t)\} = \frac{4(3s^2-4)}{(s^2+4)^3}.$$

**Answer 8.** Use the scaling property with  $c = 3$ :

$$\begin{aligned}\mathcal{L}\{f(3t)\} &= \frac{1}{3}F\left(\frac{s}{3}\right) \\ &= \frac{1}{3} \frac{1}{(s/3)^2 + 1} \\ &= \frac{1}{3} \frac{9}{s^2 + 9} \\ &= \boxed{\frac{3}{s^2 + 9}}.\end{aligned}$$

Since  $f(3t) = \sin(3t)$ , this agrees with the sine table entry.

**Worked Answers: Problems 9 To 12**

**Answer 9.** Use  $F(u) = 5/(u^2 + 25)$  in the division property:

$$\begin{aligned}
 \mathcal{L}\left\{\frac{\sin(5t)}{t}\right\} &= \int_s^\infty \frac{5}{u^2 + 25} du \\
 &= \left[\arctan\left(\frac{u}{5}\right)\right]_s^\infty \\
 &= \frac{\pi}{2} - \arctan\left(\frac{s}{5}\right) \\
 &= \boxed{\arctan\left(\frac{5}{s}\right)}, \quad s > 0.
 \end{aligned}$$

**Answer 10.** The function vanishes after  $t = 2$ , so:

$$\begin{aligned}
 \mathcal{L}\{r(t)\} &= \int_0^2 e^{-st} dt \\
 &= \left[-\frac{1}{s}e^{-st}\right]_0^2 \\
 &= \boxed{\frac{1 - e^{-2s}}{s}}
 \end{aligned}$$

for  $s \neq 0$ . At  $s = 0$ , the finite integral equals 2.

**Answer 11.** The period is  $T = 3$ . Transform one period:

$$\begin{aligned}
 \int_0^3 e^{-st} p(t) dt &= \int_0^1 2e^{-st} dt + \int_1^3 0 dt \\
 &= \frac{2(1 - e^{-s})}{s}.
 \end{aligned}$$

For Answer 11, the one-period integral is  $2(1 - e^{-s})/s$ . The periodic-transform formula divides this result by  $1 - e^{-3s}$ :

$$\boxed{\mathcal{L}\{p(t)\} = \frac{2(1 - e^{-s})}{s(1 - e^{-3s})}, \quad s > 0.}$$

**Answer 12.** Combine the kernels before identifying the shift:

$$e^{-st}e^{3t} = e^{-(s-3)t}.$$

Therefore the correct property is:

$$\boxed{\mathcal{L}\{e^{3t}f(t)\} = F(s - 3).}$$

If  $F(s)$  originally converges for  $s > \alpha$ , the shifted argument must satisfy:

$$s - 3 > \alpha.$$

Add 3 to both sides:

$$\boxed{s > \alpha + 3.}$$

## Chapter 21 Final Summary

The defining formula and the periodic-function rule are:

$$\boxed{\mathcal{L}\{f(t)\} = \int_0^{\infty} e^{-st} f(t) dt}, \quad \boxed{\mathcal{L}\{p(t)\} = \frac{\int_0^T e^{-st} p(t) dt}{1 - e^{-sT}}}.$$

The core table entries and transform-building properties are:

$$\begin{aligned} 1 &\rightarrow \frac{1}{s}, & t^n &\rightarrow \frac{n!}{s^{n+1}}, & e^{at} &\rightarrow \frac{1}{s-a}, \\ \sin(bt) &\rightarrow \frac{b}{s^2 + b^2}, & \cos(bt) &\rightarrow \frac{s}{s^2 + b^2}, \\ e^{at}f(t) &\rightarrow F(s-a), & t^n f(t) &\rightarrow (-1)^n F^{(n)}(s), & f(ct) &\rightarrow \frac{1}{c} F(s/c). \end{aligned}$$

Together, these results support one central working principle:

**identify the time-domain structure, apply one justified property at a time, and carry the convergence condition with the transform rather than treating it as an afterthought.**